UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2023

Roll No. ....

Paper: Statistical Computer Packages (Common with BS-AD 5th)
Course Code: STAT-309 Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

NOTE: USE SPSS SOFTWARE TO SOLVE YOUR QUESTIONS (WRITE DETAILED PROCEDURE OF SOLUTION IF COMPUTER IS NOT AVAILABLE DURING EXAM).

Q.1. Answer the following short questions: (6x5=30)

i) Explain how would you merge two different SPSS data files to get a single data file.

ii) Explain the restrictions on variable's naming in SPSS.

iii) If you have a variable x in SPSS, how would to transform x into a new variable $y = \sqrt{x^3}$

iv) Write the procedure of getting summary statistics (like mean, median, variance, SD, mode etc) in SPSS.

v) Write the procedure of applying an independent-samples t-test in SPSS.

vi) Write the procedure of computing spearman's rank correlation between two variables x and y.

Answers the following questions.

Q.2: During the spring semester 2020 a team of Statistics randomly sampled 35 university students to study grade point average (GPAs). The following table gives the data.

Grade Point Average

2.2	3.4	4.0	3.1	3.3	3.8	3.0
2.8	2.6	3.3	3.2	3.0	2.8	3.5
3.2	3.0	2.5	2.5	2.8	3.4	2.2
2.7	2.7	2.6	2.9	3.0	2.8	2.9
2.9	2.6	3.5	2.5	2.5	2.5	3.5

Construct and interpret 90% and 95% confidence intervals for mean. Interpret the confidence intervals. Compare the lengths of the two intervals. (08)

Q.3: Two varieties of tomato were experimented concerning their fruit-producing abilities, measures in pounds. The following data were obtained:

Location	1	2	3	4	5	6	7	8	9	10
Variety A	3.03	3.10	2.35	3.86	3.91	1.72	2.65	2.30	2.70	3.60
Variety B	2.28	3.63	2.17	3.56	3.73	1.85	1.48	1.86	2.76	2.68

Apply (i) The sign (ii) The Wilcoxon signed-rank tests to test the hypothesis that there is no difference in fruit-producing abilities of the two varieties. Use 5% level of significance. (10)

Q.4: The following data were collected which are related to the length of an infant X_1(cm), age X_2(days), and weight X_3(kg). (12)

X_1	57.5	52.8	61.3	67.0	53.5	62.7	56.2	68.5	69.2
X_2	78	69	77	88	67	80	74	94	102
X_3	2.75	2.15	4.41	5.52	3.21	4.32	2.31	4.30	3.71

a) Compute all simple correlation coefficients and interpret the results.
b) Calculate all partial correlation coefficients and interpret the results.
c) Fit a multiple linear regression line X_1 on X_2 and X_3.
d) Test the overall significance of the above regression model using ANOVA table at $\alpha = 0.05$.